

Optimal Coordinated Operation of Interdependent Power and Water Distribution Systems

Konstantinos Oikonomou^{ID}, *Member, IEEE*, and Masood Parvania^{ID}, *Senior Member, IEEE*

Abstract—With the rising electricity demand for freshwater production, power and water sectors are becoming increasingly interdependent. This vital link could create new opportunities for power and water distribution system operators to coordinate the operation of their systems and achieve operational cost and energy efficiency benefits. This paper proposes the flexible power-water flow (FlexPWF) model for co-optimizing the operation of flexible energy-intensive components in water systems with the operation of both power and water distribution systems. The FlexPWF model leverages the energy flexibility provided by water treatment and desalination plants in conjunction with variable speed pumps and water storage tanks to minimize the operating costs of interdependent power and water distribution systems. The proposed model takes into account the operational constraints of both power and water distribution systems, thus coordinating their operation without endangering the reliable supply of power and water to customers. The paper highlights the merits of integrating water treatment and desalination plants, where a detailed model is proposed to capture their dynamic operation for producing variable amounts of freshwater and providing energy flexibility. Simulations, conducted on the 33-bus test distribution system and a 15-node test water distribution system, show that the FlexPWF model effectively coordinates the operation, reduces the operating costs and improves the operational metrics of both power and water systems.

Index Terms—Interdependent critical infrastructure, flexibility, power-water nexus, distribution systems, optimization.

NOMENCLATURE

Indices and Parameters

t	Index of time intervals.
j	Index of water distribution system arcs.
n	Index of water distribution system nodes.
u	Index of variable speed pumps.
NT	Number of scheduling hours.
N	Number of water distribution nodes.
J	Number of water distribution arcs.
F	Number of freshwater reservoirs.
E	Number of wastewater reservoirs.
T	Number of water storage tanks.
U	Number of variable speed pumps.

O	Number of reverse osmosis membranes.
A	$N \times J$ node-arc incidence matrix.
B	$U \times J$ pump-arc incidence matrix.
Θ^F	$F \times N$ freshwater reservoir-node incidence matrix.
Θ^E	$E \times N$ wastewater reservoir-node incidence matrix.
Λ	$T \times N$ tank-node incidence matrix.
L	$O \times J$ membrane-arc incidence matrix.
K	$O \times O$ diagonal matrix of membrane active surface area and permeability parameters.
Π	$O \times O$ diagonal matrix of osmotic pressure coefficients and salinity parameters.
Δ	$T \times T$ diagonal matrix of tanks surface.
Z^F	$F \times 1$ vector of freshwater reservoirs elevation.
Z^E	$E \times 1$ vector of seawater reservoirs elevation.
$\bar{C}/\bar{D}$	$T \times 1$ vectors of tanks water inflow/outflow limits.
$\underline{V}/\bar{V}$	$T \times 1$ vectors of tanks min/max volume.
$\underline{Q}/\bar{Q}$	$J \times 1$ vectors of arcs min/max volumetric flow rates.
$\bar{P}^U$	$J \times 1$ vector of variable speed pumps max power consumption.
$\bar{W}^F$	$N \times 1$ vector of freshwater reservoirs max inflows.
$\bar{W}^E$	$N \times 1$ vector of wastewater reservoirs max inflows.
W_t^D	$N \times 1$ vector of freshwater demand at time t .
M_t	$N \times 1$ vector of freshwater price at time t .
Λ_t	$N \times 1$ vector of pretreatment wastewater price at time t .
λ_t^e	Forecasted wholesale energy price at power substation bus at time t .
α	Freshwater unavailability index.
a, b, c, d, e	Pump-specific performance parameters.
i, j, k	Index of power distribution system nodes.
g_i	Connected conductance at node i .
b_i	Connected susceptance at node i .
r_{ji}	Resistance of line ji .
x_{ji}	Reactance of line ji .
I_{ji}^{sq}	Square magnitude of maximum current flow limit in line ji .
$V_i^{sq} / \bar{V}_i^{sq}$	Square magnitude of min/max voltage threshold at node i .
$P_{i,t}^D$	Active power demand of node i at time t .
$Q_{i,t}^D$	Reactive power demand of node i at time t .

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The authors are with the Department of Electrical and Computer Engineering, University of Utah, Salt Lake City, UT 84112 USA (e-mail: konstantinos.oikonomou@utah.edu; masood.parvania@utah.edu).

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Decision Variables

$\mathbf{H}_t$	$N \times 1$ vector of nodal pressure heads at time t .
$\mathbf{H}_t^C$	$N \times 1$ vector of pressure heads at the exit nodes of RO membranes at time t .
$\mathbf{H}_t^S$	$N \times 1$ vector of pressure heads at the entrance nodes of RO membranes at time t .
$\mathbf{Q}_t$	$J \times 1$ vector of arcs' volumetric flow rates at time t .
$\mathbf{Q}_t^C$	$J \times 1$ vector of permeate volumetric flow rates at time t .
$\mathbf{Q}_t^S$	$J \times 1$ vector of feed volumetric flow rates at time t .
$\mathbf{W}_t^F$	$N \times 1$ vector of reservoirs freshwater inflows at time t .
$\mathbf{W}_t^E$	$N \times 1$ vector of reservoirs wastewater inflows at time t .
$\mathbf{C}_t$	$T \times 1$ vector of tanks freshwater inflows at time t .
$\mathbf{D}_t$	$T \times 1$ vector of tanks freshwater outflows at time t .
$\mathbf{V}_t$	$T \times 1$ vector of tanks stored water at time t .
$\mathbf{P}_t^U$	$U \times 1$ vector of variable speed pumps power consumption at time t .
$\mathbf{\Omega}_t$	$U \times 1$ vector of variable speed pumps relative speed at time t .
$\mathbf{Y}_t$	$O \times 1$ vector of RO membranes' freshwater recovery ratio at time t .
$\Delta \mathbf{H}_t$	$O \times 1$ vector of pressure difference across RO membranes at time t .
$\Delta \mathbf{\pi}_t$	$O \times 1$ vector of average osmotic pressure difference across RO membranes at time t .
$\hat{\Delta \pi}_{j,t}$	Linearized function of the average osmotic pressure difference at time t .
$\gamma_{j,t}^{v,m}$	Continuous decision variable associated with the sampling coordinates v and m of the linearized function at time t .
P_t^G/Q_t^G	Active/reactive power withdrawn from the upstream transmission network at time t .
$P_{ji,t}^L$	Active power flow in line ji at time t .
$Q_{ji,t}^L$	Reactive power flow in line ji at time t .
$V_{i,t}^{sq}$	Square magnitude of voltage in node i at time t .
$I_{ji,t}^{sq}$	Square magnitude of current flow in line ji at time t .
$\underline{h}_{j,t}^{v,m}$	Integer variable of lower right triangle of the linearized function at time t .
$\bar{h}_{j,t}^{v,m}$	Integer variable of upper left triangle of the linearized function at time t .

I. INTRODUCTION

A. Background and Problem Description

THE INTERDEPENDENCE between power and water systems has recently attracted the attention of both power and water system communities who seek to better understand how a synergistic operation between their systems could improve energy efficiency and reduce their operating costs [1]. This synergy is commonly referred to as the *power-water nexus* in the sense that the efficient use of one system depends on the availability and efficiency of the other [2]. Water is

needed in all power generation stages, from thermal power plant cooling to fossil fuel extraction, while power is required to treat, desalinate and distribute large amounts of high-quality water over long distances. For instance, in water-stressed states like California, water facilities consume 20% of the state's total electricity for their operations [3]. As increasing pressures of drought and population growth arise, the nexus between the power and water sector is intensified.

Water treatment and desalination are energy-intensive processes in which salt, contaminants, and other minerals are extracted to produce potable water for urban consumption, industrial use, and agricultural irrigation [4]. In the U.S., public drinking water systems supply potable water, which is first purified in water treatment facilities to more than 80% of the total population [5]. As of 2018, more than 300 municipal-level seawater and brackish water desalination plants are identified with the ability to produce at least 25,000 gallons of potable water per day in the U.S., with countless projects in the pipeline [6]. California, Florida, and Texas are the leading states in the desalination sector in U.S., both in terms of the installed capacity and the total number of operating plants [7]. In California, in drought-prone counties like San Diego, the desalinated water serves up to 8% of the freshwater demand [8]. In Florida, Tampa Bay seawater desalination plant supplies about 10% of the region's freshwater needs [9]. In Texas, there are 46 municipal brackish water desalination plants with capacities greater than 0.025 million gallons per day [10]. Desalination technologies have also been employed to process wastewater supplies. Currently, 7% of wastewater in the U.S. is reused, but only a small portion for potable uses [11].

A coordinated operation between power and water systems could uncover significant operational benefits for both infrastructures [12]. For example, electric water pumps with controllable power consumption features could be utilized by local grid operators and power utilities in the form of flexible loads to optimize the operation of both power and water distribution systems. In this regard, controllable assets from the water sector (i.e., water desalination and treatment plants, variable speed pumps), if properly managed, could reduce operating costs of both power and water systems, improve the energy efficiency of electric water pumps, reduce power and water losses, as well as offer a range of services (i.e., flexible energy, frequency regulation, congestion management) that can help grid operators counterbalance the variability and uncertainty of renewable generation resources [4], [13].

Despite the operational benefits that emerge from an integrated operation of power and water systems, yet these systems are managed, regulated, and operated independently from one another. One barrier is the concern that the optimal operation decisions of one system may not always be supported by the operational constraints of the other, which may adversely impact the primary operation objective of the critical infrastructures for reliable delivery of power and water services. For instance, although reducing the electricity consumption of pumps during periods of high energy prices would lower the operation cost of the power distribution system, it may pose a threat to reliably supply the water demand of the water

distribution system (WDS), due to insufficient pressure head gain produced by the pumps. Thus, to fully grasp the potential benefits from co-operating power and water systems, advanced operation optimization frameworks that consider both power and water operational constraints need to be developed. These frameworks should co-optimize the operation of power and water systems by leveraging the flexibility of their controllable assets and coordinate the power and water flow interactions while maintaining each system's performance objectives.

B. Literature Review

The technical literature includes multiple recent frameworks aimed at co-optimizing the operation of power and water systems. A model is proposed in [14] that integrates and optimizes the energy flexibility of WDSs' variable speed pumps (VSPs) and water storage tanks in bulk power systems operation. The work in [14] is further expanded in [15] by proposing a comprehensive model to define and optimize the feasible energy flexibility ranges of WDSs. In [16], a model is developed that co-optimizes the operation of power and water distribution systems by employing the flexibility of water storage tanks and fixed-speed pumps, and a quasi-convex hull relaxation technique is proposed to yield computationally efficient solutions. However, the use of fixed-speed pumps restricts WDS energy flexibility and limits the benefits emerging from coordinating the operation of power and water distribution systems. A model that co-optimizes the operation of power and water distribution systems by tapping on the flexible operation of water storage tanks and variable speed pumps is proposed in [17], where a feasible point pursuit successive convex approximation algorithm is adopted to obtain near-optimal solutions. A model that co-optimizes the operation of hydroelectric and desalination plants is proposed in [18], however, without considering intertemporal operation constraints and network topology of power and water systems. A model is developed in [19] for deploying the energy flexibility of water treatment plants to improve the operation of power distribution systems.

More recently, there has been significant progress in solving WDS optimization problems of different objectives by developing models based on geometric programming (GP) approaches [20]–[22]. These models convert the non-convex water flow constraints into a series of convex GP problems that can be efficiently solved by commercial solvers without prior knowledge of water flow directions or maximum flow rates. These models display several advantages in providing tractable and globally optimal solutions for large water networks, however, they are not yet implemented in power-water nexus applications. Technical literature also includes models that optimize the schedule of valves in WDSs, which could provide another degree of freedom to improve the coordinated operation of water and power distribution systems [22], [23].

In a different approach, models are proposed to integrate the energy flexibility of WDSs in power demand response (DR) programs and frequency regulation markets. These models aim

at helping water distribution system operators (W-DSOs) offset a portion of their electricity cost by providing services in power markets. A model is proposed in [24] that optimizes the schedules of VSPs to maximize W-DSO profits in the day-ahead energy market, while considering water demand uncertainties. The works in [25]–[27] develop models that leverage the flexible operation of water storage tanks, pumps, and desalination plants for maximizing W-DSO profits in both DR programs and regulation markets.

Although the technical literature includes models to co-optimize the operation of power and water systems, as well as to integrate the energy flexibility of WDSs in electricity markets, less attention has been given to investigating and modeling the operational benefits that water desalination and treatment plants could provide to the joint operation of power and water distribution systems. The models presented in [14]–[27] do not consider the flexibility provided by the dynamic operation of the water desalination and treatment plants in their co-optimization processes. Water treatment and desalination plants are energy-intensive infrastructures that can flexibly operate via controlling the amount of freshwater produced by their filtration systems. However, including the flexible operation of water plants increases the level of complexity of the power-water flow scheduling problem. Therefore, computationally-efficient mathematical models are needed that capture and schedule the energy flexibility of the water plants as part of co-optimizing the operation of power and water distribution systems.

C. Contribution and Paper Structure

This paper proposes the flexible power-water flow (FlexPWF) model for co-optimizing the operation of flexible energy-intensive components in water systems with the operation of both power and water distribution systems. More specifically, the proposed model aims to minimize the day-ahead operation costs of the interdependent power and water distribution system by leveraging the energy flexibility provided by water treatment and desalination plants, as well as by VSPs and water storage tanks. The FlexPWF model integrates the detailed operation and energy flexibility constraints of water treatment and desalination plants through a mixed-integer linear formulation that linearizes the complex nonlinear operation characteristics of water treatment and desalination plants, and takes into account their dynamic operation for producing variable amounts of freshwater via their filtration systems. In addition, the FlexPWF model enables coordinated operation of interdependent power and water distribution systems by, at one side, integrating the flexible energy consumption variables of water components in power flow constraints of the power distribution network through branch flow model, and at another side, by integrating the operational constraints of water components within hydraulic water flow constraints of the water distribution network. Therefore, the FlexPWF model enables co-optimizing the available energy flexibility without risking the reliable supply of power and water demand pertained to each system. The FlexPWF model

is formulated as a mixed-integer quadratically constrained programming (MIQCP) model, which can be efficiently solved for large-scale power and water distribution systems using commercial solvers.

The rest of this paper is organized as follows: Section II presents the proposed FlexPWF model that co-optimizes the operation of power and water distribution systems. The solution approach of the proposed model is presented in Section III. The simulation results are shown in Section IV, and the conclusions are presented in Section V.

D. Notation

The notation used in the paper to describe the power and water distribution systems is defined below. We represent the power distribution system by a directed graph $G = (\mathcal{B}, \mathcal{L})$, where $\mathcal{B} = \{1, \dots, B\}$ and $\mathcal{L} = \{ji | j, i \in \mathcal{B}, j \equiv j(i)\}$ denote the set of buses and lines, and $j(i)$ shows the buses connected to bus i . For radial systems, node $i = 1$ denotes the substation bus where the feeder is connected to the transmission system, while the distribution buses are represented by set $\mathcal{B}^b = \{2, \dots, B\}$. For each line $ji \in \mathcal{L}$, let $z_{ji} = r_{ji} + ix_{ji}$ be the complex impedance, $I_{ji,t}^{sq}$ be the squared current magnitude at time t , and $(P_{ji,t}^L + iQ_{ji,t}^L)$ be the complex power flow at time t . For each bus $i \in \mathcal{B}$, let $V_{i,t}^{sq}$ be the squared voltage magnitude at time t , $z_i = r_i + ix_i$ be the complex shunt impedance, and $y_i = 1/z_i = g_i - ib_i$ be the shunt admittance. For the substation bus, let $(P_t^G + iQ_t^G)$ be the complex power flowing from the upstream transmission system to supply the distribution system at time t . For a distribution bus, let $(P_{i,t}^{D,I} + iQ_{i,t}^{D,I})$ be the complex power of inflexible loads at time t .

Similarly, we represent the WDS by a directed graph $G = (\mathcal{N}, \mathcal{J})$, where $\mathcal{N}$ is the set of N nodes representing T tanks, F freshwater reservoirs, and E wastewater (or seawater) reservoirs; $\mathcal{J}$ is the set of J arcs comprising S pipes, U VSPs, and O reverse osmosis (RO) membranes. The pressure head of the nodes at time t is represented by the $N \times 1$ decision variable vector $\mathbf{H}_t = [H_{t,1}, \dots, H_{t,N}]^T$, and the volumetric flow rate of arcs at time t is shown by the $J \times 1$ decision variable vector $\mathbf{Q}_t = [Q_{t,1}, \dots, Q_{t,J}]^T$, where N and J respectively represent the total number of nodes and arcs. Water flow through the network is induced by different pressure levels at the nodes and is described by $\mathbf{A}^T \mathbf{H}_t$, where $\mathbf{A}$ is the $N \times J$ node-arc incidence matrix with rows and columns corresponding to nodes and arcs of the WDS, respectively.

II. FLEXIBLE POWER-WATER FLOW SCHEDULING MODEL

This section presents the proposed flexible power-water flow (FlexPWF) model. In the proposed FlexPWF model, the power and water distribution system operators (P-DSO and W-DSO) coordinate the operation of their systems to minimize day-ahead operation costs by tapping on the flexible operation of water treatment, desalination, and conveyance subsystems. These subsystems, shown in Fig. 1, consist of VSPs, water storage tanks, and filtration membranes. VSPs can operate flexibly via rapidly adjusting their speed (therefore their power consumption) to balance water load fluctuations. In water conveyance subsystems, VSPs consume electricity to generate

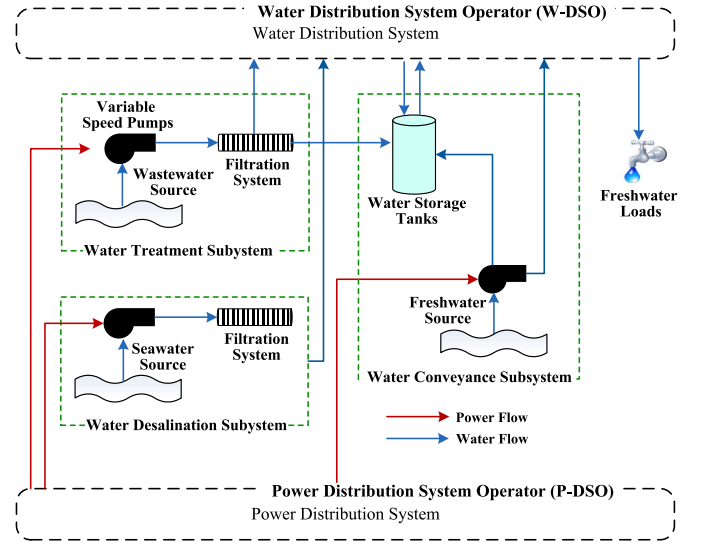


Fig. 1. Representation of interconnected power and water distribution systems.

sufficient pressure in water pipes needed to convey water over long distances, in addition to storing freshwater in the storage tanks. As the pumping energy required to deliver water to storage tanks is flexible via the utilization of VSPs, the water pumping schedule into tanks could be optimized to provide flexibility in power distribution systems. Water filtration membranes are part of both water treatment and desalination subsystems, which purify water for consumption [28]. In order to ensure freshwater productivity, VSPs are utilized on the feed side of membranes to pressurize waste or saline water. Considering the flexible operation of VSPs, water treatment and desalination plants are also candidates to provide flexibility to power distribution systems.

The proposed FlexPWF model coordinates the day-ahead flexible operation of water treatment, desalination, and conveyance subsystems in WDS with the operation of power distribution systems, assuming that both power and water flows are in steady-state while all transient states (especially those associated with the slower water flow dynamics) have damped out. In addition, we assume that appropriate intelligent metering, telecommunication, control, and automation systems at each facility and associated controllable equipment (e.g., VSPs) are in place that enable a two-way communication channel across the two systems.

The objective function of the FlexPWF model is to minimize the operation cost for meeting the electricity and water demands over the scheduling horizon of NT time intervals:

$$\begin{aligned} \min \quad & \sum_t^{NT} (P_t^G \lambda_t^e) + \sum_t^{NT} (\mathbf{1}^{UT} P_t^U \pi_t^e) \\ & + \sum_t^{NT} (\mathbf{M}_t \boldsymbol{\Theta}^F \mathbf{W}_t^F + \boldsymbol{\Lambda}_t \boldsymbol{\Theta}^E \mathbf{W}_t^E). \end{aligned} \quad (1)$$

The first term in (1) represents the cost of purchasing power P_t^G in the power distribution system, where λ_t^e is the forecasted wholesale energy price at distribution substation bus at time t . The second term in (1) is the cost the W-DSO pays to the

P-DSO to consume power, where $\mathbf{P}_t^U$ is the $U \times 1$ decision variable vector of VSPs' power consumption at time t , $\mathbf{1}^U$ is a $U \times 1$ vector of ones, and π_t^e is the energy tariff at which the W-DSO purchases energy at time t . The third term in (1) represents the cost of purchasing water from freshwater reservoirs, where $\mathbf{M}_t$ is $1 \times F$ vector of freshwater price at time t ; Θ^F is $F \times N$ freshwater reservoir-node incidence matrix; and $\mathbf{W}_t^F$ is $N \times 1$ decision variable vector of freshwater inflows at freshwater reservoir nodes at time t . The fourth term in (1) represents costs related to the wastewater and seawater pretreatment processes, where $\mathbf{\Lambda}_t$ is $1 \times E$ vector of pretreatment wastewater price at time t ; Θ^E is $E \times N$ wastewater reservoir-node incidence matrix; and $\mathbf{W}_t^E$ is $N \times 1$ decision variable vector of wastewater inflows at wastewater reservoir nodes at time t . The objective function (1) is constrained to the operation constraints of power and water distribution networks, which are presented next.

A. Power Distribution System Model

The power flow in power distribution systems is modeled using the branch flow formulation, which accounts for both active and reactive power flows and integrates the active and reactive power of flexible WDS pumping loads in the distribution buses [29]. The active and reactive power flows in the substation bus are formulated in (2) and (3). The active and reactive power balance constraints in distribution buses that integrate WDS flexible pumping loads are formulated in (4) and (5), where $(P_{u,i,t}^U + iQ_{u,i,t}^U)$ is the complex power consumption of a variable speed pump $u \in \mathcal{U}$ at time t . The voltage drop constraint is formulated in (6) in terms of the active and reactive power flows of the lines. The bus voltage limits are enforced in (7). Constraint (8) limits the complex power flows of the lines, while the current flows are constrained to the maximum current limit of lines in (9). The inequality constraint (8) represents a convex form of the original equality constraint $V_{i,t}^{sq} I_{i,t}^{sq} = P_{i,j}^2 + Q_{i,j}^2$ which is relaxed as a rotated second order cone constraint.

$$P_t^G = \sum_{1k \in \mathcal{L}} P_{1k,t}^L + g_1 V_{1,t}^{sq}, \quad \forall t, \quad (2)$$

$$Q_t^G = \sum_{1k \in \mathcal{L}} Q_{1k,t}^L + b_1 V_{1,t}^{sq}, \quad \forall t, \quad (3)$$

$$-P_{i,t}^{D,I} - \sum_{u \in \mathcal{U}_i} P_{u,i,t}^U = \sum_{ik \in \mathcal{L}} P_{ik,t}^L - \sum_{ji \in \mathcal{L}} (P_{ji,t}^L - r_{ji} I_{ji,t}^{sq}) + g_i V_{i,t}^{sq}, \quad i \in \mathcal{B}^b, \forall t, \quad (4)$$

$$-Q_{i,t}^{D,I} - \sum_{u \in \mathcal{U}_i} Q_{u,i,t}^U = \sum_{ik \in \mathcal{L}} Q_{ik,t}^L - \sum_{ji \in \mathcal{L}} (Q_{ji,t}^L - x_{ji} I_{ji,t}^{sq}) + b_i V_{i,t}^{sq}, \quad i \in \mathcal{B}^b, \forall t, \quad (5)$$

$$V_{i,t}^{sq} - V_{j,t}^{sq} = -2(r_{ji} P_{ji,t}^L + x_{ji} Q_{ji,t}^L) + (r_{ji}^2 + x_{ji}^2) I_{ji,t}^{sq}, \quad (ji) \in \mathcal{L}, \forall t, \quad (6)$$

$$\underline{V}_i^{sq} \leq V_{i,t}^{sq} \leq \overline{V}_i^{sq}, \quad i \in \mathcal{B}^b, \forall t, \quad (7)$$

$$V_{i,t}^{sq} I_{ji,t}^{sq} \geq P_{ji,t}^2 + Q_{ji,t}^2, \quad (ji) \in \mathcal{L}, \forall t, \quad (8)$$

$$I_{ji,t}^{sq} \leq \overline{I}_{ji}^2, \quad (ji) \in \mathcal{L}, \forall t. \quad (9)$$

B. Water Distribution System Model

The WDS delivers freshwater from the freshwater reservoirs to the water consumers, as well as seawater from the seawater reservoirs to the water treatment plants via a system of pipes, water storage tanks, and variable speed pumps.

1) *Mass Balance Constraint*: The mass balance constraint for WDS nodes is given in (10), where $\mathbf{W}_t^D$ is $N \times 1$ vector of nodes' freshwater demand parameter at time t ; $\mathbf{C}_t$ and $\mathbf{D}_t$ are respectively $T \times 1$ vectors of freshwater inflow and outflow decision variables of water storage tanks at time t ; and $\mathbf{\Lambda}$ is $T \times N$ tank-node incidence matrix. The freshwater inflow and outflow variables of tanks are included in (10), allowing W-DSOs to leverage the water storage capacity of tanks to store freshwater and supply freshwater demand at a later time. The freshwater and wastewater inflows are respectively limited in (11) and (12), where $\overline{\mathbf{W}}^F$ and $\overline{\mathbf{W}}^E$ are the maximum freshwater and wastewater inflow bounds. Note that the wastewater inflow first passes through a RO filtration system, which separates dissolved salts and other minerals before supplying the freshwater demand. Thus, only a portion of the wastewater inflow is eventually converted to freshwater, based on the design characteristics of the RO membrane. In addition, based on the availability of freshwater resources and the possibility of facing freshwater shortages, the W-DSO may choose to recover higher portions of freshwater from the water treatment or desalination plants. This is mathematically expressed in (13), where $\mathbf{Y}_t \in [0, 1]$ is $O \times 1$ vector of the membranes' freshwater recovery ratio at time t ; $\mathbf{L}$ is $O \times J$ membrane-arc incidence matrix; $\alpha \in [0, 1]$ is the freshwater unavailability index that remains fixed throughout the scheduling horizon; and $\mathbf{1}^N$ is a $1 \times N$ row vector of ones; in (13), higher values for α indicate that a higher portion of the freshwater demand is supplied from the water treatment or desalination plants.

$$\mathbf{W}_t^E + \mathbf{W}_t^F - \mathbf{W}_t^D = \mathbf{A} \mathbf{Q}_t + \mathbf{\Lambda}^T [\mathbf{C}_t - \mathbf{D}_t], \quad \forall t, \quad (10)$$

$$0 \leq \mathbf{W}_t^F \leq \overline{\mathbf{W}}^F, \quad \forall t, \quad (11)$$

$$0 \leq \mathbf{W}_t^E \leq \overline{\mathbf{W}}^E, \quad \forall t, \quad (12)$$

$$\mathbf{1}^N \mathbf{Y}_t^T \mathbf{L}^T \mathbf{W}_t^E \geq \alpha \mathbf{1}^N \mathbf{W}_t^D, \quad \forall t. \quad (13)$$

2) *Water Pipes*: The head loss in pipes is modeled by the empirical *Hazen-Williams* constraint (14), where $\mathbf{G}$ is $J \times J$ diagonal matrix of pipes' resistance parameters, and $\Pi(\cdot)$ is the unit sign function [30]. The nodal pressure heads are confined in (15) to the minimum and maximum pressure head bounds at the nodes. The nodal pressure heads are respectively fixed at the freshwater and wastewater reservoir nodes in (16) and (17) by a fixed elevation height, where $\mathbf{Z}^F$ and $\mathbf{Z}^E$ are respectively $F \times 1$ and $E \times 1$ vectors that contain freshwater and wastewater reservoir nodes' elevation data; The volumetric flow rate variable of arcs is limited in (18), where the bound $\overline{\mathbf{Q}}$ depends on the pipes' and pumps' design characteristics.

$$-\mathbf{A}^T \mathbf{H}_t = \mathbf{G} \Pi(\mathbf{Q}_t) |\mathbf{Q}_t|^{1.852}, \quad \forall t, \quad (14)$$

$$\underline{\mathbf{H}} \leq \mathbf{H}_t \leq \overline{\mathbf{H}}, \quad \forall t, \quad (15)$$

$$\Theta^F \mathbf{H}_t = \mathbf{Z}^F, \quad \forall t, \quad (16)$$

$$\Theta^E \mathbf{H}_t = \mathbf{Z}^E, \quad \forall t, \quad (17)$$

$$-\overline{\mathbf{Q}} \leq \mathbf{Q}_t \leq \overline{\mathbf{Q}}, \quad \forall t. \quad (18)$$

3) *Water Treatment and Desalination Model*: Water treatment and desalination plants provide an alternative to freshwater reservoir sources for supplying freshwater. As shown in Fig. 1, water treatment and desalination plants consist of a high pressure pump and a RO membrane filtration system. Reverse osmosis is the dominant technology for mid- and large-size water treatment and desalination plants as it provides a cost effective option of producing high quality freshwater from seawater or wastewater with lower electricity consumption requirements than other technologies [28]. In this process, the pressurized wastewater (or seawater) stream (feed) is split into a freshwater stream (permeate) and a concentrated brine stream (retentate) [31], [32].

The pressures and flow rates for the RO membrane are presented in (19)–(23). The permeate flow rate $\mathbf{Q}_t^C \subset \mathbf{Q}_t$ is modeled in (19), where $\Delta \mathbf{H}_t$ is $O \times 1$ vector of pressure difference across the membrane (transmembrane pressure); $\Delta \pi_t$ is $O \times 1$ vector of the average osmotic pressure difference between the retentate and permeate streams along the membrane; $\mathbf{K}$ is $O \times O$ diagonal matrix of membranes' active surface area and hydraulic permeability parameters; and $\mathbf{1}^O$ is a $O \times 1$ vector of ones. In (20), the pressure difference across the membrane is equal to the difference between the pressure of the feed water at the entrance of the membrane $\mathbf{H}_t^S \subset \mathbf{H}_t$ and the pressure of the permeate water at the exit of the membrane $\mathbf{H}_t^C \subset \mathbf{H}_t$. The average osmotic pressure difference is approximated in (21), where the membrane's freshwater recovery ratio $\mathbf{Y}_t$ is equal to the ratio of permeate water flow rate to the feed water flow rate $\mathbf{Q}_t^S \subset \mathbf{Q}_t$ in (22); Π is $O \times O$ diagonal matrix of the osmotic pressure coefficients of the feed water. In order to ensure permeate productivity along the membrane, the transmembrane pressure has to overcome the average osmotic pressure difference, as imposed in (23).

From a modeling perspective, water treatment plants are modeled in the same way as water desalination plants based on the assumption that both plants use the same type of filtration technology (i.e., reverse osmosis) to respectively filter seawater and wastewater. However, the distinction between the two plants is made by considering different salinity coefficients for seawater and wastewater which impact the amount of energy used to purify the water. The different levels of saline concentration associated with the feed water are represented by the parameter Π in constraint (21).

$$\mathbf{Q}_t^C = \mathbf{L}^T \mathbf{K} (\Delta \mathbf{H}_t - \Delta \pi_t), \quad \forall t, \quad (19)$$

$$\Delta \mathbf{H}_t = \mathbf{L} \mathbf{A}^T (\mathbf{H}_t^S - \mathbf{H}_t^C), \quad \forall t, \quad (20)$$

$$\Delta \pi_t = \Pi (\mathbf{1}^O \oslash (\mathbf{1}^O - \mathbf{Y}_t)), \quad \forall t, \quad (21)$$

$$\mathbf{Y}_t = \mathbf{L} (\mathbf{Q}_t^C \oslash \mathbf{Q}_t^S), \quad \forall t, \quad (22)$$

$$\Delta \mathbf{H}_t \geq \Delta \pi_t, \quad \forall t. \quad (23)$$

4) *Water Storage Tanks*: Let $\mathbf{V}_t$ denote the $T \times 1$ vector of water volume in tanks at time t . The state equation that describes the dynamic operation of water storage tanks is given in (24). The water volume of tanks is confined to the minimum and maximum volume limits in (25). The water volume stored in tanks at the beginning of scheduling horizon is initialized

in (26), where $\mathbf{V}^{init}$ is the vector of initial water volume stored in tanks. In (27), the water volume varies based on the pressure head at the tank nodes, where Δ is $T \times T$ diagonal matrix of tanks' surface values. Constraints (28) and (29) respectively limit tanks' water inflow and outflow rates to the maximum inflow and outflow rate bounds.

$$\mathbf{V}_{t+1} = \mathbf{V}_t + (\mathbf{C}_t - \mathbf{D}_t) \Delta_t, \quad \forall t, \quad (24)$$

$$\underline{\mathbf{V}} \leq \mathbf{V}_t \leq \bar{\mathbf{V}}, \quad \forall t, \quad (25)$$

$$\mathbf{V}_1 = \mathbf{V}^{init}, \quad (26)$$

$$\mathbf{V}_t = \Delta \mathbf{A} \mathbf{H}_t, \quad \forall t, \quad (27)$$

$$0 \leq \mathbf{C}_t \leq \bar{\mathbf{C}}, \quad \forall t, \quad (28)$$

$$0 \leq \mathbf{D}_t \leq \bar{\mathbf{D}}, \quad \forall t. \quad (29)$$

5) *Variable Speed Pumps*: The pressure head increase in pipes generated by VSPs and their power consumption are respectively formulated in (30) and (31), where Ω_t is $U \times 1$ decision variable vector of VSPs relative speed at time t ; $\mathbf{B}$ is $U \times J$ pump-arc incidence matrix; $\mathbf{1}^U$ is a $U \times 1$ vector of ones; and a, b, c, d and e are pump-specific parameters [33]. Signs $\odot$ and $\oslash$ in (30) and (31) refer respectively to component-wise matrix multiplication and division. In (32), $U \times 1$ decision variable vector of VSPs' power consumption, $\mathbf{P}_t^U$, is constrained to its maximum power capacity, $\bar{\mathbf{P}}^U$, while the water flow rate through a pump takes positive values in (33), denoting that VSPs can operate only in one direction.

$$\mathbf{B} \mathbf{A}^T \mathbf{H}_t = \Omega_t^2 \odot (a \mathbf{1}^U - b((\mathbf{1}^U \oslash \Omega_t) \odot \mathbf{B} \mathbf{Q}_t)^c), \quad \forall t, \quad (30)$$

$$\mathbf{P}_t^U = \Omega_t^3 \odot (d \mathbf{1}^U - e((\mathbf{1}^U \oslash \Omega_t) \odot \mathbf{B} \mathbf{Q}_t)), \quad \forall t, \quad (31)$$

$$0 \leq \mathbf{P}_t^U \leq \bar{\mathbf{P}}^U, \quad \forall t, \quad (32)$$

$$\mathbf{B} \mathbf{Q}_t \geq 0, \quad \forall t. \quad (33)$$

In summary, (1)–(33) presents the formulation of the proposed FlexPWF model, which is solved using a MIQCP reformulation next.

III. SOLUTION APPROACH

The proposed FlexPWF model is a non-convex and non-linear programming (NLP) problem with nonlinearities represented by the pressure head loss constraint (14), the VSPs operation constraints (30) and (31), the water treatment or desalination plants operation constraints (21) and (22), as well as the complex power flow quadratic constraint (8). NLP models, however, can be computationally intensive for large WDSs, and a feasible global optimal solution may not always be guaranteed. In our previous works [15], [26], we developed models to approximate and replace the nonlinear functions of constraints (14), (30) and (31) with convex piecewise linear alternatives, and showed that linear approximations lead to fairly accurate solutions in much lower computation times. The readers are referred to [15], [26] for linear models for constraints (14), (30) and (31). In this paper, we present a model that utilizes continuous and binary variables to linearize water treatment and desalination constraint (21), in which the freshwater recovery ratio $\mathbf{Y}_t$ is substituted by (22).

A. Linear Water Treatment and Desalination Constraints

Let function $\Delta\pi_{j,t}(Q_{j,t}^C, Q_{j,t}^S) = \pi_{n,t} - \pi_{n+1,t}$ be the time-dependent osmotic pressure difference across the RO membrane connecting two consecutive nodes n and $n+1$, where $Q_{j,t}^C$ and $Q_{j,t}^S$ are the permeate and feed water flow rates at time t , respectively. Using (20), $\Delta\pi_{j,t}(Q_{j,t}^C, Q_{j,t}^S)$ is a nonlinear function defined as $\Delta\pi_{j,t}(Q_{j,t}^C, Q_{j,t}^S) = \Pi(1/(1 - (Q_{j,t}^C/Q_{j,t}^S)))$. In order to develop the piecewise linear approximation of the bivariate function $\Delta\pi_{j,t}(Q_{j,t}^C, Q_{j,t}^S)$, we divide the axis $Q_{j,t}^C$ and $Q_{j,t}^S$ respectively to $\mathcal{V} - 1$ and $M - 1$ segments using breakpoints $k_j^1, k_j^2, \dots, k_j^{\mathcal{V}}$ and $w_j^1, w_j^2, \dots, w_j^M$. The values k_j^1 and $k_j^{\mathcal{V}}$ as well as w_j^1 and w_j^M coincide with the lower and upper bounds of the $Q_{j,t}^C$ and $Q_{j,t}^S$ domains [34]. The segments on the axes partition the bivariate function domain into several rectangles. Consider the rectangle defined by vertices $(k_j^v, w_j^m), (k_j^{v+1}, w_j^m), (k_j^{v+1}, w_j^{m+1}), (k_j^v, w_j^{m+1})$ on the function domain. This rectangle is covered by two triangles defined by vertices $(k_j^v, w_j^m), (k_j^v, w_j^{m+1}), (k_j^{v+1}, w_j^{m+1})$, and $(k_j^v, w_j^m), (k_j^{v+1}, w_j^m), (k_j^{v+1}, w_j^{m+1})$, and divided by the diagonal $(k_j^v, w_j^m), (k_j^{v+1}, w_j^{m+1})$. A continuous variable $\gamma_{j,t}^{v,m} \in [0, 1]$ is associated with each vertex (v, m) , and binary variables $\bar{h}_{j,t}^{v,m}$ ($v = 1, \dots, \mathcal{V} - 1; m = 2, \dots, M$) and $\underline{h}_{j,t}^{v,m}$ ($v = 2, \dots, \mathcal{V}; m = 1, \dots, M - 1$) are respectively associated with the upper left and lower right triangles.

Any given operating point $k_j^v \leq Q_{j,t}^C \leq k_j^{v+1}$ and $w_j^m \leq Q_{j,t}^S \leq w_j^{m+1}$ can be uniquely represented by (34) and (35), as linear combinations of the vertex coordinates weighted by the associated variables $\gamma_{j,t}^{v,m}$. The bivariate nonlinear function $\Delta\pi_{j,t}(Q_{j,t}^C, Q_{j,t}^S)$ is then approximated by the linear function $\hat{\Delta}\pi_{j,t}$ in (36), as convex combination of the function values evaluated at the vertices of the triangle containing the operating point $(Q_{j,t}^C, Q_{j,t}^S)$, weighted by the associated variables $\gamma_{j,t}^{v,m}$. Constraint (37) ensures that in any solution only one of the triangles is chosen, while (38)-(40) ensure that in any solution at most three $\gamma_{j,t}^{v,m}$ can take non-zero values.

$$Q_{j,t}^C = \sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M k_j^v \gamma_{j,t}^{v,m}, \quad \forall t, \forall j, \quad (34)$$

$$Q_{j,t}^S = \sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M w_j^m \gamma_{j,t}^{v,m}, \quad \forall t, \forall j, \quad (35)$$

$$\hat{\Delta}\pi_{j,t} = \sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M \Delta\pi_{j,t}(k_j^v, w_j^m) \gamma_{j,t}^{v,m}, \quad \forall t, \forall j, \quad (36)$$

$$\sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M (\bar{h}_{j,t}^{v,m} + \underline{h}_{j,t}^{v,m}) = 1, \quad \forall t, \forall j, \quad (37)$$

$$\sum_{v=1}^{\mathcal{V}} \sum_{m=1}^M \gamma_{j,t}^{v,m} = 1, \quad \forall t, \forall j, \quad (38)$$

$$\gamma_{j,t}^{v,m} \leq \bar{h}_{j,t}^{v,m-1} + \bar{h}_{j,t}^{v+1,m} + \bar{h}_{j,t}^{v,m}, \\ + \underline{h}_{j,t}^{v-1,m} + \underline{h}_{j,t}^{v,m+1} + \underline{h}_{j,t}^{v,m}, \quad \forall t, \forall j, \forall v, \forall m, \quad (39)$$

$$0 \leq \gamma_{j,t}^{v,m} \leq 1, \quad \forall t, \forall j, \forall v, \forall m. \quad (40)$$

The linear model in (34)-(40) approximates the nonlinear functions in (21) and (22), and the linear models

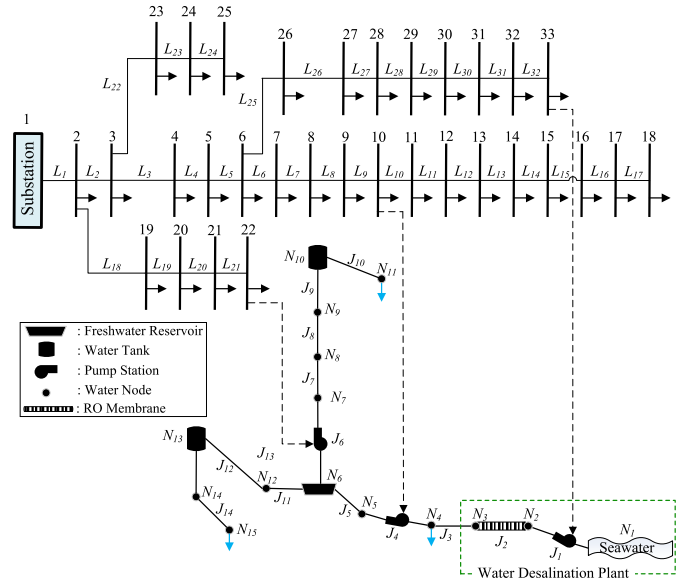


Fig. 2. Interdependent 33-bus test power distribution and 15-node water distribution systems [35].

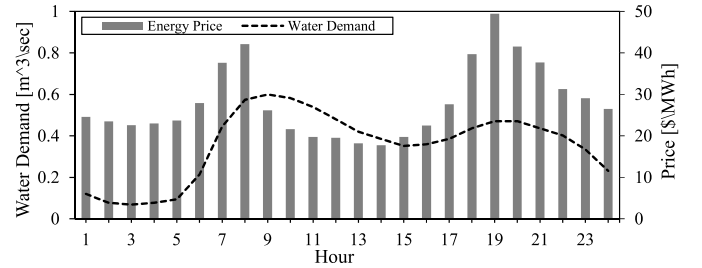


Fig. 3. Hourly forecasted energy price and water demand profile.

introduced in [15], [26] approximate the nonlinear functions in (14), (30) and (31). In addition, the relaxed branch flow model is used to formulate the AC power flow constraints (2)-(9) consists of a set of linear and quadratic constraints. Substituting the nonlinear functions with the associated linear approximations render the proposed FlexPWF model into a convex MIQCP model, which is solvable as a whole using computationally tractable MIQCP solvers, e.g., CPLEX.

IV. SIMULATION RESULTS

The proposed FlexPWF model is implemented on the 33-bus distribution test system that supplies the electricity of a 15-node WDS [35]. The topology of the interdependent power and water distribution systems is shown in Fig. 2. In Fig. 2, the WDS consists of 15 nodes, 14 arcs (including 10 pipes, 3 variable speed pumps in arcs 1, 4, 6, and 1 RO membrane in arc 2), two water storage tanks (nodes 10, 13), three water load nodes (nodes 4, 11, 15), one freshwater reservoir (node 6) and one wastewater reservoir (node 1). The saline concentration associated with the wastewater is set to 30 kg/m^2 (at 25°C). The California's forecasted residential water demand profile, shown in Fig. 3, is used as water demand in the simulation [36]. The full data for the test WDS can be found in [35].

In addition, we assume that the freshwater unavailability index α is 0.5, implying that at least 50% of the hourly

TABLE I
DAILY ENERGY COSTS

	P-DSO Cost (\$)	W-DSO Cost (\$)	Total Cost (\$)
Case 1	3,484.3	1,105.0	4,589.3
Case 2	3,423.4	1,041.5	4,464.9
Case 3	3,345.3	940.8	4,286.1

freshwater demand is supplied from the water treatment plant. The hourly load data of California Independent System Operator (CAISO) for March 30, 2018 is scaled down to the 33-bus test system's active and reactive peak loads of 3,715kW and 2,300kVar, and used in the simulations. In addition, CAISO's wholesale energy prices of the same day are used in the simulations [37]. The upper and lower limits of the voltage magnitude at each distribution bus are set to 0.9 p.u. and 1.04 p.u. The three VSPs are supplied by buses 10, 22, and 33, respectively. It is assumed that in buses with electric water pumps, the W-DSO pays a fixed energy tariff of 0.11\$/kWh to the power distribution system operator for consuming electricity.

The numerical results are provided for three case studies. In Case 1, the W-DSO operates its WDS without coordination with the power distribution network. In this case, the W-DSO solves an operation problem that minimizes the WDS operating costs subject only to the WDS hydraulic constraints (10)-(33). The W-DSO operating costs consist of the pumping energy cost as well as wastewater pre-treatment and water purchasing costs. The pumping energy cost is related to the energy tariff that the W-DSO pays to the P-DSO to consume energy. The tariff could have any shape depending on the rate structure of the P-DSO. The W-DSO then passes the optimized WDS energy consumption schedules to the P-DSO who minimizes its energy costs subject to the power distribution network constraints (2)-(9). In Case 2, the operation of power and water distribution systems is coordinated using the FlexPWF model proposed in Section II, and the freshwater recovery ratio Y_t of the water desalination plant is fixed to 0.3. Case 3 is similar to Case 2, however, Case 3 can exploit the flexible operation of desalination plant, where the freshwater recovery ratio varies between 0.1 and 0.5. The simulation results are presented next.

1) *Daily Energy Costs*: The daily energy costs of P-DSO and W-DSO to respectively operate the 33-bus power distribution system and the 15-node water distribution system in the three case studies is shown in Table I. In Table I, the total energy cost of both systems is reduced to \$4,464.9 in Case 2 as compared to Case 1, which demonstrates the operational cost benefits of coordinating the operation of two systems. The total daily energy cost in Case 3, where the water desalination plant operates in varying freshwater recovery ratios, is further reduced to \$4,286.1, showing that the flexible operation of water desalination plant impacts the daily energy costs of both systems, meanwhile supplying power and water to the respective customers. Overall, Case 3 demonstrates a total of 6.6% energy cost savings as compared to Case 1 based on a single-day operation.

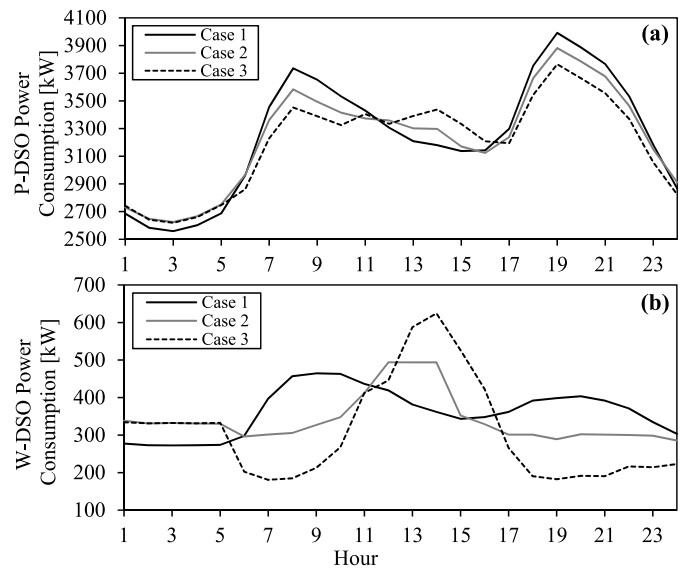


Fig. 4. Active power consumption schedules of (a) P-DSO, (b) W-DSO.

2) *Power Consumption Schedules*: The hourly active power consumption schedules of the power and water distribution systems are shown in Figs. 4. In Fig. 4(a), the active power consumption of the power distribution system in Cases 2 and 3 drops during high energy price hours 6-10 and 17-21, and increases during low energy price hours 1-5 and 11-16, as compared to Case 1. This result shows that the coordinated operation of the two systems in Cases 2 and 3 has smoothed out the daily load profile and reduced the peak load of the 33-bus system as compared to Case 1. This is more salient in Case 3 where the flexible operation of water desalination plant offers additional operational flexibility to the P-DSO.

In Fig. 4(b), the W-DSO active power consumption profile in Case 1 follows the same trend as the water demand profile shown in Fig. 3. In Cases 2 and 3, however, coordinated operation of the two systems modifies the electricity consumption profile of the W-DSO as compared to Case 1. More specifically, in Case 3, the W-DSO reduces the power consumption of the pumps during high energy price periods 6-10 and 17-21 and shifts the reduced load to lower energy price periods 1-5 and 11-16.

This operational flexibility emerges from coordinating the operation of VSPs with the flexible operation of water desalination plant and water storage tanks in the WDS. In Fig. 5(a), the seawater feed pump operates during hours 1-5 and 11-16 in Case 3 to recover higher portions of freshwater from the water desalination plant to supply the freshwater demand of the WDS. During the same hours, the rest of the VSPs operate to fill the water tanks with freshwater, as shown in Fig. 5(b). On the other hand, during hours 6-10 and 17-21, power consumption of VSPs is reduced, as the freshwater demand is supplied by the water storage tanks and consequently less freshwater is produced from the desalination plant. In addition, as shown Fig. 5(b) in Cases 1 and 2, less water is stored in the tanks and subsequently released from them over the scheduling horizon as compared to Case 3. This result shows that the flexible operation of the desalination plant allows the W-DSO to more effectively utilize the storage capacity of the

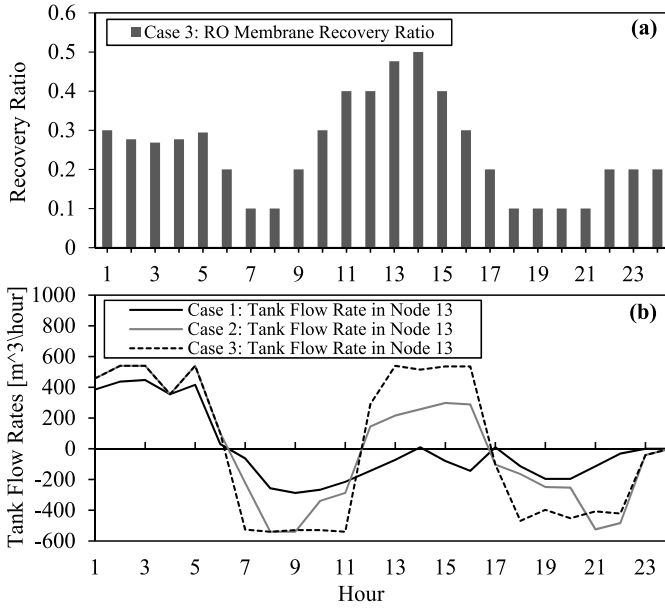


Fig. 5. (a) Desalination plant freshwater recovery ratio, (b) tank flow rate.

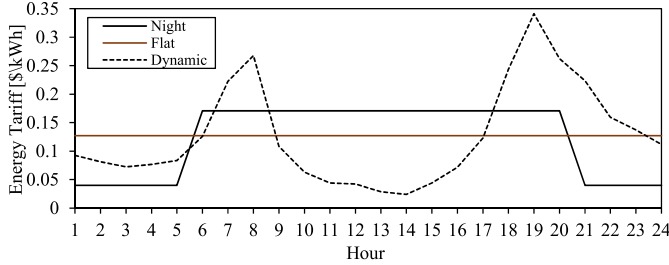


Fig. 6. Energy tariff applied to water distribution system.

tanks via better coordinating the operation of VSPs with water tanks' charge and discharge schedules.

A. Sensitivity Analysis

Three sensitivity analyses are conducted to evaluate the impacts of retail energy tariff, water demand profile, and freshwater availability on the operational benefits and energy flexibility that controllable WDS assets can provide to the coupled power and water distribution system.

1) *Impact of WDS Energy Tariff*: In this study, we aim at investigating the impacts of different energy tariff schemes on the WDS scheduling results. The energy tariff, π_t^e , used to minimize the W-DSO energy costs in Case 1 was considered flat over the scheduling horizon and equal to 0.11\$/kWh. However, it is often common for the P-DSOs to offer time-varying energy tariffs to the W-DSOs for purchasing electricity. In this regard, we repeat the simulations for Cases 1 and 3 using two additional WDS energy tariff schemes, namely *night* and *dynamic*, shown in Fig. 6. In the night scheme, the W-DSO minimizes its energy costs by taking advantage of the low WDS energy tariffs during night hours. In the dynamic scheme, the WDS energy tariff follows the trend of the wholesale energy prices.

The simulation results are provided in Fig. 7 and Table II. In Fig. 7, the flat scheme provides little incentives to the W-DSO in Case 1 to alter the operation of its VSPs, resulting in a

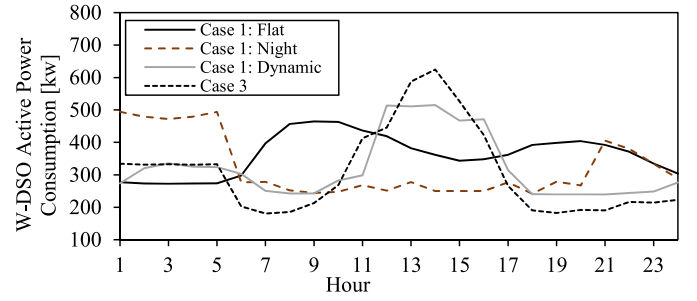


Fig. 7. W-DSO active power consumption under different tariff schemes.

TABLE II
DAILY W-DSO ENERGY COST

Tariff Scheme	W-DSO Cost (\$)	
	Case 1	Case 3
Flat	1,105.0	940.8
Night	979.0	935.8
Dynamic	955.2	735.3

TABLE III
IMPACT OF FRESHWATER UNAVAILABILITY IN CASE 3

Freshwater Unavailability Index (%)	DSO Cost (\$)	W-DSO Cost (\$)	Total Cost (\$)	Percentage Change (%)
30	3,326.1	769.8	4,095.9	-4.4
50	3,345.3	940.8	4,286.1	0
70	3,380.9	1,059.4	4,440.3	3.5

power consumption schedule which simply follows the water demand profile. In the night scheme, the W-DSO operates the VSPs in Case 1 to leverage the low energy tariffs during night hours 1-5 and 21-24 to recover higher portions of freshwater from the water treatment plant as well as store water in the tanks. In the dynamic scheme, the W-DSO in Case 1 takes advantage of the low energy tariffs during hours 12-16, which results in significantly increasing its electricity consumption during these hours and decreasing it during the remaining ones. It is worth highlighting, that neither the night nor the dynamic energy tariff schemes can alter the WDS power consumption schedule in Case 1 to the extent it is optimized in Case 3. This is shown in Table II, where Case 3 records lower daily W-DSO energy costs as compared to Case 1 in all three different energy tariff schemes. This result shows that the coordinated operation of the two systems in Case 3 leads to larger cost savings for the W-DSO and increases WDS energy flexibility as compared to the uncoordinated Case 1, even when incentive-based tariff schemes are present.

2) *Impact of Freshwater Unavailability*: The goal of this study is to assess the impact of freshwater unavailability index α in (13) on the results. To this end, we repeat the simulations for Case 3 above (where α equals to 50%), using freshwater unavailability values of 30% and 70%. The simulation results are provided in Table III, where the total operation cost of both power and water distribution systems increases as the freshwater unavailability index increases. For high freshwater

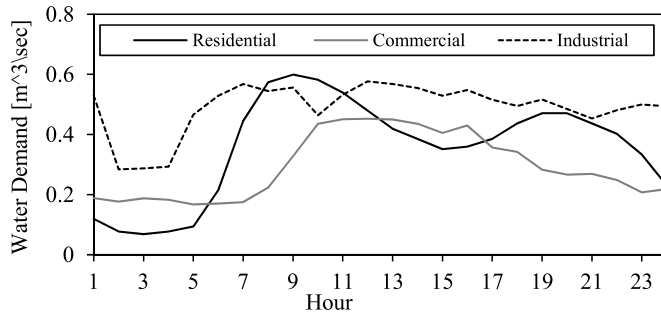


Fig. 8. Water demand profiles for sensitivity analysis 3.

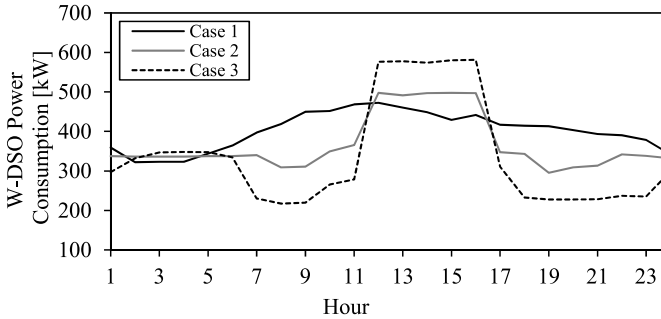


Fig. 9. W-DSO power consumption under different water demand profiles.

unavailability value of $\alpha = 70\%$, a higher portion of the freshwater demand has to be supplied by the energy-intensive water desalination facility, leading to a 3.5% increase of the total operation cost as compared to the total operation cost for lower values of α . On the other hand, when there is a more diversified freshwater water supply portfolio with $\alpha = 30\%$, the W-DSO more effectively coordinates the water desalination process with the power distribution system operation, leading to 4.4% saving in the total operation cost. This result shows that the unavailability of freshwater, which in turn increases the W-DSO's dependence on the desalination plant, limits the available flexibility that is coordinated between the interconnected power and water systems.

3) *Impact of Water Demand Profile*: In this study, we assign different water demand profiles to nodes 4, 11, 15 of the WDS, in order to reflect different freshwater consumption sectors on the coordinated operation of the systems. The water demand profiles for node 4 (residential), node 11 (commercial), and node 15 (industrial), adopted from [38], are illustrated in Fig. 8. The simulation results performed to compute the P-DSO and W-DSO costs for the three cases in Table I, are repeated here using the water demand profile information in Fig. 8. The daily energy costs for the two systems are summarized in Table IV, and the W-DSO active power consumption profiles for the three cases are shown in Fig. 9. The change in water demand profile increases the operation cost of both water and power systems in uncoordinated Case 1. The operation costs are reduced in Table IV as the operation of the systems is coordinated in Cases 2 and 3. In Case 3, where the water desalination plant can produce varying amounts of freshwater during the scheduling horizon, a higher reduction of the total operation cost is observed in Table IV as compared to Case 2. In addition, as shown in Fig. 9, the energy intensive process

TABLE IV
DAILY ENERGY COSTS IN SENSITIVITY ANALYSIS 3

	DSO Cost (\$)	W-DSO Cost (\$)	Total Cost (\$)
Case 1	3,673.9	1,233.6	4,907.5
Case 2	3,616.8	1,118.2	4,735.0
Case 3	3,592.3	1,000.3	4,592.6

of water desalination in Cases 2 and 3 are shifted to low electricity price hours (11-17), demonstrating similar scheduling decisions as those displayed in Fig. 4.(b). This result shows that the nodal water demand profiles affect the operation of both power and water system, and the proposed model flexibly coordinate the schedules of the WDS controllable assets with power distribution system operations given the time-varying water demand profiles.

B. Computation Time

The computation time for solving the FlexPWF model in Cases 2 and 3 is presented here. The two case studies are solved using CPLEX 12.6.2 on a computer with a 3.6 GHz i7 CPU and 64GB of RAM, where the optimality gap is set to zero. The optimal solutions for Cases 2 and 3 are respectively obtained after 4.15 and 5.93 seconds. As expected, in Case 3, the operational benefits that emerge from the flexible operation of the desalination plant come with a slight increase in computation time. Nevertheless, in both case studies the solution of the FlexPWF model is obtained fairly fast, demonstrating efficiency of the proposed formulation and promise of applicability to larger systems.

V. CONCLUSION

This paper proposed the FlexPWF model that co-optimizes the operation of power and water distribution systems by leveraging the energy flexibility provided by water treatment and desalination plants as well as the variable speed pumps and water tanks in order to minimize the operation costs of both systems. The paper emphasizes on the additional operational benefits and energy flexibility that water treatment and desalination plants could provide to interdependent power and water distribution systems and a mixed-integer linear model is proposed to capture its flexible operation for providing variable amounts of freshwater.

Simulation results of implementing the model on the 33-bus test power distribution system and a 15-node water distribution system show that the coordinated operation between power and water distribution systems reduces their operation costs, especially in the case where water desalination plants operate at varying freshwater recovery ratios. Further, the flexible operation of water desalination plants allows W-DSOs to better utilize the water storage capacity of tanks as the additional energy flexibility improves the coordination of electric water pumps with water storage tanks' charge and discharge schedules. Multiple sensitivity analyses are conducted showing that dynamic energy tariff schemes lead to larger cost savings for the W-DSO as compared to traditional night tariff incentives. In addition, we showed that the proposed model co-optimizes

the operation of the two interdependent systems under different freshwater availability conditions, where energy flexibility is balanced between securing the freshwater demand and minimizing operation costs. Moreover, the proposed model can coordinate the flexibility across the two systems considering different water demand profiles.

Future work includes expanding the proposed model to a stochastic optimization framework, which would account for the uncertainty of water and power demand forecasts. In addition, extension of the proposed model to integrate various types of valves in coordinated operation of water and power distribution systems is an important topic that may be studied in future works. The development of cybersecurity tools and protocols to enable secure implementation of the proposed coordinated operation framework is also in order.

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Konstantinos Oikonomou (Member, IEEE) received the B.Sc. degree in physics from the University of Patras, Patras, Greece, in 2009, the first M.Sc. degree in energy engineering from the University of Durham, U.K., in 2011, the second M.Sc. degree in electrical engineering from the George Washington University, USA, in 2014, and the Ph.D. degree in electrical engineering from the University of Utah, USA, in 2019. His research interests include power system planning and reliability analysis as well as operation and planning of interdependent critical infrastructures, water-energy systems, and modeling and integration of renewable energy resources.

Masood Parvania (Senior Member, IEEE) is currently an Associate Professor and an Associate Chair of the Department of Electrical and Computer Engineering, and the Director of Utah Smart Energy Laboratory, University of Utah. His research interests include the operation, economics and resilience of power and energy systems, and modeling and operation of interdependent critical infrastructures. He serves as an Associate Editor of the IEEE TRANSACTIONS ON SMART GRID and the IEEE POWER ENGINEERING LETTERS, and is the Chair of IEEE Power and Energy Society (PES) Utah Chapter, the Chair of IEEE PES Bulk Power Systems Operation Subcommittee, and the Vice-Chair of IEEE PES Risk, Reliability, and Probability Applications Subcommittee.